

OligoTox Open Data Challenge — Phase 2

Methodology Document

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Submission target:	Single PDF, ≤5 pages, 8.5"×11", ≥1" margins, ≥11pt Arial, line spacing ≥1.0

STATUS: DRAFT. Materials and methods below describe the protocol as designed. Before submission, confirm each method was actually followed as written (or update to reflect any deviations made during real data generation) — the challenge requires this document to describe "the materials and methods used to generate the data contained in the dataset files," i.e., what was actually done, not only what was planned.

1. Oligonucleotide Library

Library spans 7 backbone chemistries (PS, PO, PMO, LNA-mix, PNA, 2'-F-ANA, morpholino), 4 sugar modifications (DNA, 2'-OMe, 2'-F, LNA/mixed), and multiple structural classes (gapmers, siRNA/siRNA-like duplexes, aptamer scaffolds, splice-switching oligos, GalNAc- and lipid-conjugated ASOs), plus systematic CpG-content, GC-content, and length gradients, known clinical/reference compounds, and scrambled/non-targeting negative controls.

2. Oligonucleotide Synthesis, Purification, and Identity Characterization

(This section directly answers the challenge's required content: "methods used to purify and characterize oligo identity.")

- **Synthesis:** Solid-phase phosphoramidite synthesis via CRO partner(s) [name vendor(s) actually used, e.g., IDT / Eurofins Genomics].
- **Purification:** Reverse-phase HPLC (RP-HPLC) purification for all oligos; target purity ≥95% for standard chemistries, ≥90% for heavily modified chemistries.
- **Identity confirmation:** Mass spectrometry (LC-MS) confirmation of molecular weight against expected mass for each synthesized sequence/chemistry combination.
- **Purity re-assessment on outliers:** Any compound producing an unexpected toxicity signal is re-confirmed by MS before inclusion in the final dataset.
- **Endotoxin testing:** LAL or equivalent assay; acceptance threshold <1 EU/mL in working solution, to rule out endotoxin contamination as a confound in immunotoxicity/PBMC readouts.

DRAFT — PENDING REAL EXPERIMENTAL DATA

- **Formulation:** Sterile filtration, lyophilization, and reconstitution in RNase-free PBS prior to dosing.
- **Reporting per compound:** vendor, synthesis scale, HPLC purity (%), MS-confirmed mass (pass/fail), endotoxin level (EU/mL) — all recorded in the Compound Registry table of the dataset (see `dataset/data_dictionary_and_schema.md`).

3. In Vitro Human Model Systems

[List only the systems actually used to generate the submitted data; delete unused systems from the planned five below or annotate as "not yet run."]

1. **Primary human hepatocytes (PHH), sandwich culture** — hepatotoxicity panel.
2. **Human kidney proximal tubule organoids** — nephrotoxicity panel.
3. **Human PBMC co-culture** — immunotoxicity and complement activation panel.
4. **Human platelet-rich plasma / citrated plasma** — thrombocytopenia and coagulopathy panel.
5. **Liver-kidney microphysiological system (organ-on-chip)** — multi-organ / chronic exposure panel.

For each system used, report: cell/tissue source and donor count, dosing concentrations and timepoints actually tested, and replicate structure (biological × technical).

4. Toxicology Readouts

Each system's readouts (viability, membrane integrity, apoptosis markers, organ-injury biomarkers, cytokine/complement multiplex, platelet activation/aggregation, coagulation timing assays, and — for subsets — transcriptomic/proteomic profiling) are enumerated in the endpoint registry referenced in the dataset files. Report which of the planned endpoints were actually measured for the submitted data.

5. Dosing and Replication

Report actual concentrations/doses tested per system, actual number of biological replicates (donors/preparations) and technical replicates per condition, and any anchor/reference compounds repeated across batches for drift monitoring.

6. Quality Control

- **Plate-level:** Z'-factor and vehicle-control CV computed per plate; plates failing threshold repeated.

- **Compound-level:** Outlier detection (e.g., Grubbs test) across replicates; dose-response curve fit quality (R^2) flagged when below threshold.
- **Batch-level:** Anchor compounds repeated across batches; inter-batch CV monitored, with normalization applied if drift detected.
- Report actual QC pass rates and any compounds/plates excluded, with justification.

7. Data Processing and Curve Fitting

Raw instrument output normalized to vehicle control; dose-response curves fit via 4-parameter logistic regression to derive IC₅₀/EC₅₀ (with 95% CI), E_{max}, Hill coefficient, NOEC, and AUDRC. Software/package and version used for fitting should be named here for reproducibility.

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